Project Proposal

AI-Powered Smart Tourism Assistant Platform with Personalized Trip Planning

CS3202 – Software Engineering Project

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Group 10

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1 Overview of the Project

The proposed project aims to develop an AI-powered smart tourism assistant platform that simplifies and enhances the travel planning process through a centralized mobile and web application.

Currently, tourists need to use multiple applications and websites for accommodation, restaurants, attractions, weather information, and local events. This makes travel planning fragmented, time-consuming, and inefficient.

To address this issue, the proposed system will provide a single integrated platform where users can interact with an AI-powered assistant through a conversational interface and receive personalized travel recommendations and optimized itineraries based on their preferences, budget, travel duration, and interests.

The system will include a mobile application, web application, backend services, and an administrative dashboard for managing tourism-related data such as hotels, restaurants, attractions, and local events.

The proposed system will accept various user inputs including travel preferences and interests, budget constraints, travel duration, destination or district, number of travellers, and preferred travel style. Based on these inputs, the system will generate personalized itineraries and provide attraction recommendations, hotel and restaurant suggestions, budget estimations, nearby attraction information, and optimized travel schedules tailored to the user's requirements.

2 Objectives of the Project

- Design and implement an AI-powered smart tourism assistant platform.
- Provide personalized trip planning and recommendation services.
- Develop a conversational interface for travel assistance.
- Automate itinerary generation based on user preferences and constraints.
- Provide budget planning and optimization for travellers.
- Develop an administrative dashboard for tourism content management.
- Deliver both mobile and web applications for improved accessibility.

3 The Need for the Project

Tourists often rely on multiple applications and websites to obtain information about destinations, hotels, restaurants, attractions, weather conditions, and local events. This

fragmented approach makes travel planning time-consuming and inconvenient, particularly for users who are unfamiliar with a destination.

The proposed smart tourism assistant addresses these challenges by providing a centralized platform that generates personalized recommendations and optimized itineraries through an AI-powered conversational interface. By integrating tourism data, budget planning, and route optimization into a single system, the platform aims to improve convenience, reduce planning effort, and enhance the overall travel experience.

4 Scope of the Project

The proposed system aims to develop an AI-powered smart tourism assistant platform that supports personalized trip planning through web and mobile applications. The system will recommend attractions, hotels, restaurants, nearby places, and travel schedules based on user preferences and budget.

4.1 Traveler / General User

Users can register and log into the system, enter travel details such as destination, budget, travel duration, interests, and travel style, and interact with the AI assistant through a conversational interface. The system will generate personalized itineraries and recommend suitable attractions, hotels, and restaurants. Users will also be able to view nearby places and manage their travel plans.

4.2 Administrator

The administrator will manage tourism-related information such as attractions, hotels, restaurants, and local events through an admin dashboard. The administrator will also monitor system usage and reports.

4.3 AI Planning Module

The AI planning module will analyze user requirements and generate personalized recommendations and travel itineraries based on budget, interests, travel duration, and user preferences.

5 Deliverables

- Mobile application.
- Web application.

- AI-powered itinerary generation and recommendation engine.
- Administrative dashboard.
- Centralized tourism database.
- RESTful APIs.

6 Overview of Existing Systems and Technology

6.1 Existing Similar Systems

- **Mindtrip**: An AI-powered travel planning platform that generates personalized itineraries through a conversational interface. It integrates interactive maps and allows users to organize their trips in a single platform [1].
- **Layla**: An AI travel assistant that generates travel itineraries and helps users find hotels and flights. The system redirects users to external booking platforms to complete reservations [2].
- **Google Travel**: A travel planning platform that helps users organize destinations, accommodation, flights, and travel reservations in one place. It also provides destination information and trip management features [3].
- **Wanderlog**: A trip planning and itinerary management platform that allows users to organize travel plans, collaborate with travel companions, and manage reservations across multiple devices [4].

Although existing travel planning platforms provide useful features, several limitations still exist:

- Most platforms depend heavily on external booking services and focus mainly on hotels and transportation rather than complete trip planning.
- Existing systems often rely on publicly available information and may not provide curated or locally verified tourism data.
- Many platforms provide itinerary suggestions but require users to manually adjust plans to satisfy practical constraints such as time, budget, and travel distance.

The proposed system addresses these limitations by providing a centralized AI-powered tourism platform that combines personalized recommendations, budget planning, itinerary generation, and route optimization within a single application. The system will utilize curated tourism data and generate travel plans based on user preferences, budget, travel duration, and interests.

6.2 Technology Stack

Component	Technology
Frontend (Web)	React.js + Tailwind CSS
Backend	Python
Backend API	FastAPI
Large Language Model	Gemini API
Database	PostgreSQL
ORM	SQLAlchemy
Authentication	JWT
Maps & Navigation	Google Maps Platform
External Services	Weather API, Google Places API
Version Control	Git + GitHub
Containerization	Docker

References

[1] Mindtrip, "Mindtrip," [Online]. Available: `https://mindtrip.ai/`. [Accessed: Jul. 4, 2026].

[2] Layla AI, "Layla AI," [Online]. Available: `https://layla.ai/`. [Accessed: Jul. 4, 2026].

[3] Google, "Google Travel," [Online]. Available: `https://www.google.com/travel/`. [Accessed: Jul. 4, 2026].

[4] Wanderlog, "Wanderlog," [Online]. Available: `https://wanderlog.com/`. [Accessed: Jul. 4, 2026].

[5] "AI-Driven Smart Tourism System: Enhancing Travel Experiences Through Intelligent Automation," ResearchGate, 2025. [Online]. Available: `https://www.researchgate.net/publication/391477675_AI-Driven_Smart_Tourism_System_Enhancing_Travel_Experiences_Through_Intelligent_Automation`. [Accessed: Jul. 5, 2026].

[6] "AI-Powered Smart Tourism Assistant Platform," IEEE Xplore, Art. no. 11032743, 2025. [Online]. Available: `https://ieeexplore.ieee.org/document/11032743/`. [Accessed: Jul. 5, 2026].